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① M-tap casual moving average filter

Definition:

$$h[n] = \frac{1}{M} \sum_{k=0}^{M-1} \delta[n-k]$$

Freq. response (take DTFT)

$$H(e^{jw}) = \frac{1}{M} \sum_{k=0}^{M-1} e^{-jwk}$$

This is a geometric series:

$$H(e^{jw}) = \frac{1}{M} \sum_{k=0}^{M-1} e^{-jwk} \cdot \frac{1 - e^{-jwM}}{1 - e^{-jw}}$$

Factoring out to get magnitude & phase

$$H(e^{jw}) = \frac{1}{M} \cdot e^{-jw(M-1)/2} \cdot \frac{\sin(wM/2)}{\sin(w/2)}$$

Magnitude:

$$|H(e^{jw})| = \frac{1}{M} \left| \frac{\sin(wM/2)}{\sin(w/2)} \right|$$

Phase: $\angle H(e^{jw}) = -\frac{w(M-1)}{2}$

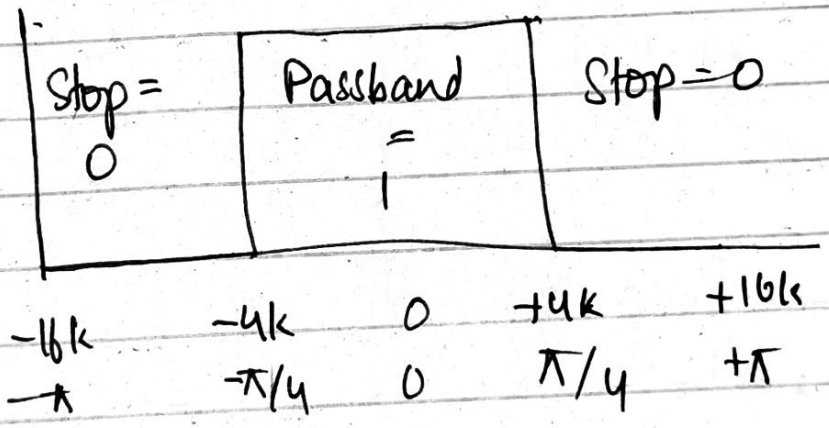
Characteristics:

- low pass filter
- linear phase
- magnitude = 1 at $\omega = 0$, zeros at $\omega = 2\pi k/M$.

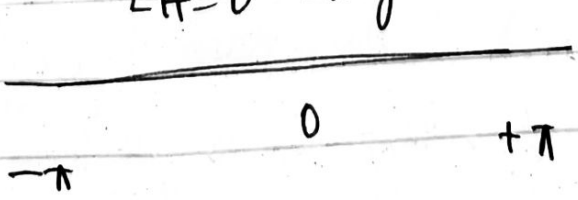
(B2) Ideal LPF & HPF:

$f_c = 4kHz$, $f_s = 32kHz$

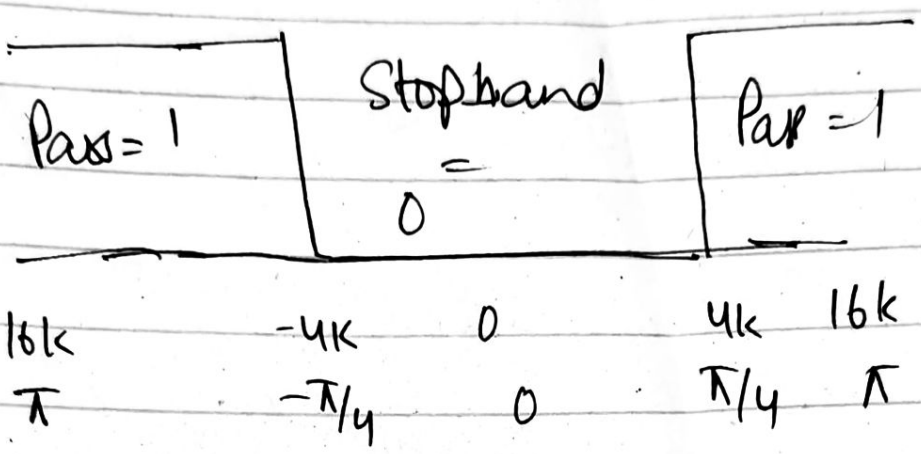
Ideal LPF - magnitude



Ideal LPF - phase



Ideal HPF



Ideal HPF - phase

$\angle H = 0$ everywhere (0 phase)

(B3) Report:

Fixed 300 Hz was too conservative and rumble's harmonics went well over 300 Hz. 700 Hz was chosen were rumble's energy was more decayed (not completely inexistent tho).